

RAGUL B

AI Engineer | Data Science Graduate

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PROFILE SUMMARY

AI Engineer with hands-on experience integrating LLM APIs and deploying RAG-backed applications end-to-end — covering API orchestration, backend architecture, and cloud deployment via FastAPI, Streamlit, and Render. Extended into classical ML pipelines with PySpark. Strong Python foundation. AWS Certified Cloud Practitioner.

EXPERIENCE

Data Analytics Intern

Jun 2025 - Aug 2025

Share Market Profile — Coimbatore, India

- Built and maintained Python-based data pipelines processing 10,000+ structured financial records, applying preprocessing and validation logic reusable across downstream reporting systems.
- Reduced data quality defects by ~25% through systematic cleansing and validation of financial datasets.
- Built Power BI dashboards to translate technical outputs into stakeholder-facing reporting for a 4-member team.
- Synthesised exploratory findings into concise briefings within Agile sprint cycles.

PROJECTS

RD AI – Multi-Modal GenAI Assistant | Python, Streamlit, Groq (LLaMA), SQLite, RAG

- Integrated the Groq LLaMA 3.3-70B API and LLaMA 4 Scout vision model into a production Streamlit app, orchestrating a 4-source RAG pipeline (DuckDuckGo, Wikipedia, and document sources) with confidence scoring and 7-intent detection.
- Engineered multi-format document ingestion (PDF, DOCX, XLSX, CSV) and persistent SQLite session storage; deployed and maintained the app on Streamlit Cloud.
- Designed a confidence-scoring mechanism across the RAG pipeline to improve response reliability, and built a custom floating UI component for production-grade UX.

MODUS – AI Transformation Strategy Intelligence | Streamlit, FastAPI, Google Gemini, SQLAlchemy

- Architected a FastAPI backend integrating the Google Gemini API via an OpenAI-compatible endpoint, with SQLAlchemy/SQLite persistence and a DuckDuckGo-based research layer.
- Deployed the backend on Render and the frontend on Streamlit Cloud as separate services, managing a two-service production architecture.
- Handled Render free-tier cold-start behavior (30-60s) with graceful frontend loading states, ensuring reliable service handoff between the two deployments.

ODI AI Team Selector | Python, PySpark, Scikit-learn, FastAPI

- Deployed a composite ML scoring model, trained on 50,000+ historical ODI records, as a FastAPI service — presented as a research paper at INFUSE 2025, Jain University.
- Engineered 10+ performance-driven features using PySpark to enable reproducible model training and evaluation at scale.

TECHNICAL SKILLS

LLM APIs & Integration: OpenAI API, Groq API, Google Gemini API, Prompt Engineering, Agentic Workflow Design

RAG & Retrieval: RAG Pipeline Design, Multi-source Keyword Retrieval (DuckDuckGo, Wikipedia), Multi-source Retrieval Orchestration

Backend & Deployment: FastAPI, Streamlit, SQLAlchemy, Render, Streamlit Cloud

Machine Learning: PySpark, Scikit-learn, Feature Engineering, Predictive Modelling

Programming: Python (Pandas, NumPy, Scikit-learn), SQL

Cloud & Tools: AWS Certified Cloud Practitioner, Git, GitHub, Jupyter Notebook

Soft Skills: Analytical Thinking, Decision Making, Communication, Attention to Detail, Time Management

EDUCATION

M.Sc. Data Science and Analytics — Jain (Deemed-to-be University), Bengaluru

2024 - 2026

CGPA: 7.3 / 10

B.Sc. Computer Technology — Hindusthan College, Coimbatore

2021 - 2024

CGPA: 7.9 / 10

CERTIFICATIONS

AWS Certified Cloud Practitioner | IBM Data Science Professional Certificate | INFUSE 2025 International Conference — Paper Presentation, Jain University